Elie Jerdak

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Tel: (+961) 76 03 33 23 **Portfolio** https://portfolio-5doo.onrender.com/

Summary

Computer Science graduate and backend-focused Full-Stack Developer experienced in building web applications and REST APIs using ASP.NET Core, Node.js, and modern frontend technologies. Strong foundation in backend architecture, authentication and authorization (JWT, RBAC), and object-oriented programming. Experienced in developing AI-powered applications, including a full-stack Retrieval-Augmented Generation (RAG) system integrating document processing, vector search, and LLM-based answer generation. Proficient in Python for machine learning and data analysis, with a focus on building secure, maintainable, and scalable software solutions.

Education

- **Bachelor of Science in Computer Science**
  Université Saint Esprit Kaslik (USEK) — Graduated with Honors, 2026

Experience

- **Software Engineering Intern | MT2**
  June 2025 – July 2025
  - Designed and implemented a full-stack Customer Management System (CMS) for gym facilities using ASP.NET Core and Entity Framework Core, focusing on scalable backend architecture and modular service design.
  - Built a secure authentication system combining JWT and HTTP-only cookies to support API security and session-based MVC access control.
  - Integrated Stripe API for automated payment processing and subscription handling, including backend webhook handling for payment confirmation.
  - Developed a QR-based check-in system for members, implementing secure token generation and validation for controlled access workflows.
  - Deployed the Fitness Club Management System on Render using Docker-based deployment architecture: **https://fitnessclub-cms.onrender.com/**

Projects

- **Lexis — RAG System | ASP.NET Core, Next.js, TypeScript, Python, Docling, PyMuPDF, Ollama, Qdrant, PostgreSQL, Redis, Cloudflare R2**
  - Developed a full-stack RAG application with a Next.js frontend and ASP.NET Core backend, enabling users to upload documents and ask questions grounded in their content.
  - Built a document-processing pipeline using Docling and PyMuPDF for PDF extraction, structure-aware semantic chunking, table-continuation handling, and source-page provenance.
  - Implemented document embeddings using Ollama (`nomic-embed-text`), Qdrant vector storage, and hybrid retrieval combining dense semantic search with lexical search.
  - Integrated LLM-based answer generation grounded in retrieved document chunks, with persistent source citations linked to document chunks and conversation messages.
  - Implemented JWT authentication with refresh tokens, PostgreSQL persistence, Redis caching, Cloudflare R2 object storage, and API rate limiting.

- **Car Price Predictor & SoFIFA Data Analysis | Python, Django, Machine Learning**
    - Built a machine learning regression pipeline in Python to predict car prices, deployed through a Django-based web interface for interactive user input and predictions.
    - Performed data preprocessing and feature engineering on the SoFIFA dataset, applying unsupervised anomaly detection techniques including Local Outlier Factor (LOF), k-Nearest Neighbors (kNN) distance-based detection, and PCA-based reconstruction error to identify and handle abnormal data points.
    - Applied SHAP (Shapley Additive Explanations) to interpret model predictions and analyze feature influence on output behavior.
    - Deployed the application publicly on Render: **https://used-car-price-predictor-mr54.onrender.com/**

- **Course Management System | Node.js, Express, EJS**
    - Developed a server-rendered Course Management System using Node.js, Express, and EJS following the MVC architecture, implementing full CRUD functionality for courses, users, and enrollments.
    - Designed a modular backend structure separating concerns across authentication, course management, and administrative operations to improve maintainability and scalability.
    - Implemented server-side rendering to dynamically generate views from backend state without reliance on frontend frameworks.
    - Deployed the application on Render for public access and testing: **https://course-management-1g9y.onrender.com/**

- **Connect-4 AI Engine | Minimax, Alpha-Beta Pruning**
    - Implemented an AI-powered Connect-4 game using the Minimax algorithm optimized with Alpha-Beta pruning for efficient decision-making.
    - Built a simulation system for AI vs AI gameplay to evaluate strategy performance and optimize move selection logic.
    - Developed a visual evaluation system to demonstrate AI decision paths and game state analysis.
    - Deployed the project publicly using GitHub Pages: **https://elie-jerdak.github.io/connect4-ai/**

Skills

- Languages: C#, Java, C++, JavaScript (Node.js), Python, SQL, HTML, CSS
- Backend & Frameworks: ASP.NET Core MVC, Node.js, Express, FastAPI
- Databases: SQL Server, MySQL, MongoDB
- Machine Learning & Data: Scikit-learn, Feature Engineering, SHAP, Unsupervised Anomaly Detection
- Backend & System Design: REST APIs, JWT Authentication, MVC Architecture, OOP Principles
- Tools & Platforms: Git, Stripe API, Power BI
- CS Fundamentals: Data Structures & Algorithms

Languages

- Arabic: Native
- English: Fluent (Professional Working Proficiency)
- French: Working Proficiency

Note:

- **The deployed applications may take a few seconds to start because they are hosted on Render's free tier.**

- **Please use the following credentials should you like to test them:**

    - **Fitness-Club:**

        - **Email:** jardakelie@gmail.com

        - **Password:** Password_123

    - **Course Management System**

        - **Admin :**

            - **Email**: elie@email.com
            - **Password**: Password_123

        - **Instructor:**

            - **Email**: philippe@email.com
            - **Password**: Password_123

        - **Student:**

            - **Email:** maroun@email.com
            - **Password**: Password_123